TELLUS



PRODUCT VERSIONS

This technical data sheet is related to following version of Tellus OTS dashboard:

	TELLUS OTS
PN#	6100666/XX
Power supply	12 Vdc
Contents	
Inputs/outputs	✓
CAN bus 1, 2	✓
LIN bus	✓
LEDs bar graph	✓
2 analog indicators	✓
25 LED telltales	✓
2 push buttons	✓
Buzzer	✓
Certifications	
Type approval ECE ONU reg10-05 E3[1]	✓
MTA Studio ID[2]	0001

GENERAL DESCRIPTION

Tellus is a dashboard designed for agricultural or off-highway vehicles, on wheels or tracks.
Power supply is 12 Vdc.

Tellus common features:
- 1 color 3.5" QVGA TFT, brightness min 400 cd/m^2;
- 1 buzzer;
- 25 telltales;
- 28 digital inputs;
- 4 analog inputs;
- 2 frequency inputs;
- 2 tristate inputs;
- 1 D+ (alternator);
- 3 communication lines (2 CAN, 1 LIN);
- 4 outputs (high side drivers);
- 2 push buttons;
- 3 LED bar graphs.

Tellus can be used as a dashboard with the following features:
- hour meter, odometer, gear, speed, engine RPM, trip data, etc.;
- PTO symbols, PTO RPM, clock, temperature, voltmeter;
- errors (FMI - SPN), service, etc.;
- analog signal interface and monitor;
- digital signal from external sensors.

A real time clock is available to give timing information, without backup battery.

It features a memory for permanent data storage (EEPROM).

Tellus is configurable with MTA Studio and MTA Gate[3].

MTA STUDIO

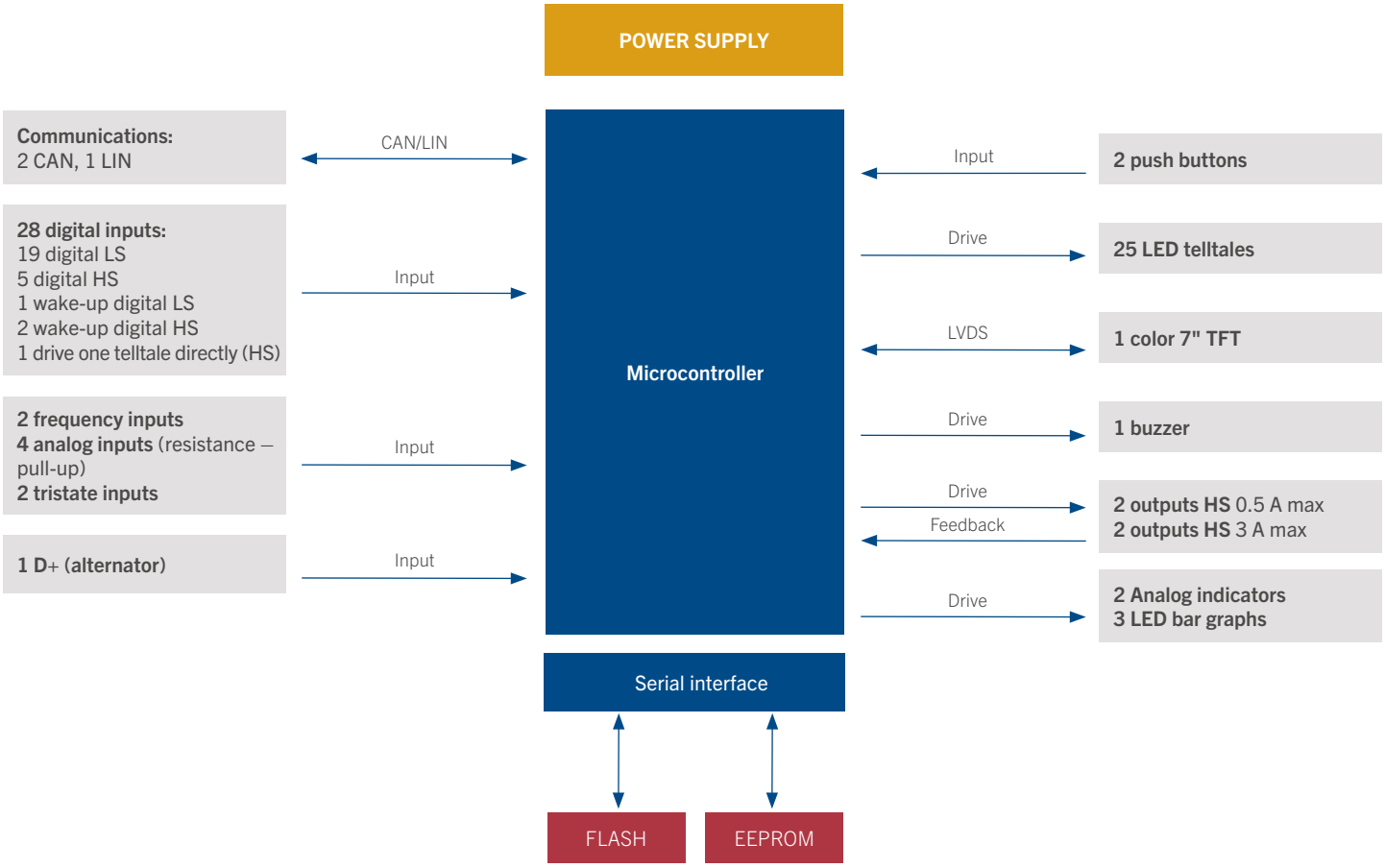
MTA Studio is a specific tool used to program dashboards, displays and electronic control units which have been developed with customizable functions allowing the customers to create "tailored" devices.

MTA GATE

MTA Gate is a specific tool designed for EOL department and/or for technical service points use.

1. Extension of ECE ONU reg. 10-05 will be done after the product qualification.
2. The "MTA Studio ID" here listed are referred to the MTA Studio variants, for cluster TELLUS product family, written on the product label. Please contact MTA for any info about it.
3. Please contact MTA to verify which features can be programmed with MTA Studio and MTA Gate. Custom development could be evaluated by MTA in case of specific functionalities needed and not already available.

BLOCK DIAGRAM



SPECIFICATIONS

CHARACTERISTIC	DESCRIPTION	QTY
Operating voltage		
Nominal	12 Vdc	
Supply voltage, permissible range	9÷16 Vdc	
Operating temperature		
External ambient temperature	-30 to +75 °C	
Storage temperature		
External ambient temperature	-40 to +85 °C	
Protection rate		
IP65		
Weight		
700 g		
Mounting location (ref. TS 50/16)		
Cab interior C1		
Telltales		**25**
High efficiency LED monochrome	Light intensity driven by PWM	25
Functionality	Telltales driven by microcontroller	24
	Telltales driven by microcontroller and by one digital input directly	1
Display		**1**
TFT 3.5" QVGA, FFS technology	Color, 240xRGBx320, 24-bits digital RGB	1

Microcontroller		
ARM Cortex M7	Cypress Traveo II CYT3DL 2.5D graphics supported	
Clock frequency		
MHz	240 (PLL0)	
Internal memory capacity in microcontroller		
Flash	4160 KB	
RAM	512 KB	
Video RAM	2048 KB	
EEPROM	Emulated	
Memories		
Flash	32 MB	
PCB temperature sensor		
Internal	NTC sensor, accuracy: ±5%	
Battery voltage measurement		
Internal	Accuracy: ±5% of full scale value	
Backlight		
LCD	White with dimming function	
Interfaces		**3**
CAN bus (up to 1 Mb/s)	ISO 11898	2
LIN bus	LIN 2.1 Master	1

CHARACTERISTIC	DESCRIPTION	QTY
CAN bus termination resistor		
Termination resistor	CAN1 available on a dedicated pin CAN2 not provided	
Current consumption		
Without load, 12 Vdc at 25 °C	350 mA at key-on 230 µA at key-off	
Analog inputs		**4**
Resistance mode	12 bit	4
Digital inputs		**28**
Active low	Active low	19
	With wake-up capability	1
Active high	Active high	5
	With wake-up capability	2
	With direct drive of one telltale	1
Frequency inputs		**2**
Alternator (W) inputs	Up to 5 kHz	2
Alternator input (D+)		**1**
Digital input read as analog 9-32 Vdc	12 bit	1
Internal load	56 Ω	
Minimal excitation current	0.16 A	
High side outputs		**4**
Digital output with current sense	Max. 0.5 A	2
Digital output with current sense	Max. 3 A	2
Buzzer		**1**
Internal piezo buzzer	SPLmin: 78 dB(A) ±6 dB(A), 2.0 kHz, 10 cm	

Fault detection in the event of cable break and short circuit		
All connector signals (inputs, outputs, CAN lines) are protected against permanent short circuit to ground and VBat	Maximum short-circuit voltage: 18 Note: for the short circuit protection it's necessary to apply one MiniVal fuse with 7.5A on the battery pin and one on the key pin	
Reverse polarity protection		
Power supply	✓ Maximum polarity inversion is -14 Vdc	
Inputs	✓ NOTE: For the short circuit protection it's necessary to apply one MiniVal fuse with 7.5A on the battery pin and one on the key pin	
Outputs	✓	
Wake-up		
From key and digital input pin 22 of CN1 and pin 8, 9 of CN2		
Push buttons		**2**
Configurable functions in MTA Studio	Rubber, not illuminated	2
Materials and other mechanical characteristics		
Front mask	PC+ABS PA6 GF25 black	1
Back cover	PC+ABS PA6 GF25 black	1
Keypad	Black silicon rubber 40 shA	1
Protective glass features		
Cover glass	PMMA 8N with antifog treatment	1

CHARACTERISTICS

DIGITAL, ANALOG AND TRISTATE INPUTS

Characteristics

- Digital active low inputs (pins 1..7, 9..13, 16..19, 22..25 of CN1):
  - 0-VBat voltage range;
  - all digital inputs are read as analog channel;
  - the contact resistance must be ≤10 Ω;
  - pin 22 of CN1 with wakeup capability;
  - switch detection threshold:
    - active input: (VLmax) = 3.2 Vdc;
    - open input: (VHmin) = 4.8 Vdc.
- Digital active high inputs (pin 26 of CN1; pin 7..12, 17 of CN2):
  - 0-VBat voltage range;
  - all digital inputs are read as analog channel;
  - the contact resistance must be ≤10 Ω;
  - pin 8 and 9 of CN2 with wakeup capability;
  - pin 26 of CN1 drive one telltale directly;
  - switch detection threshold:
    - active input: (VHmin) = 4.8 Vdc;
    - open input: (VLmax) = 3.2 Vdc.
- Tristate inputs (pins 22 and 23 of CN2):
  - 0-VBat voltage range;
  - all tristate inputs are read as analog channel;
  - the contact resistance must be ≤ 10 Ω;
  - switch detection threshold:
    - high status: (VLmax) = 4.8 Vdc;
    - low status: (VHmin) = 3.2 Vdc;
    - open status: VLmin = 3 Vdc and VHmax = 4.8 Vdc.
- Analog input (pins 14, 18, 20 and 21 of CN2):
  - pin 14: pull-up to Vref 270 Ω resistor; accuracy: 12% of the read value in the range 5 Ω to 290 Ω;
  - pin 18: pull-up to Vref 2.7K Ω resistor; accuracy: 10% of the read value in the range 500 Ω to 2 KΩ;
  - pin 20: pull-up to Vref 820 KΩ resistor; accuracy: 15% of the read value in the range 37 Ω to 1.15 KΩ;
  - pin 21: pull-up to Vref 270 Ω resistor; accuracy: 10% of the read value in the range 97 Ω to 172 Ω.

Wake-up functionality

The digital inputs pin 22 of CN1, pin 8 and 9 of CN2 have the wake-up functionality.

Wetting current activation

All digital inputs provide cleaning contact function always active by a wetting current value equal to 2.0 mA at 12 Vdc.

FREQUENCY INPUT

The signal applicable into the input (pin 24 and 25 of CN2) can be defined in the following modes:

- alternator (W) type signal (square wave);
- maximum expected frequency: 5 KHz (min 5 Hz), duty cycle typ. 50% (min 30%, max 70%);
- voltage range of the input signal: 0-VBat;
- frequency measurement accuracy: ± 10% of read value.

OUTPUTS

500 mA high side output

- 2 high side output 500 mA (pin 5 and 6 of CN2):
  - On/off mode;
  - iLoad: 500 mA;
  - Imax: 1 A;
  - minimum load resistance (Rload): min 27 Ω at14 Vdc (32 Ω at 16 Vdc);
  - power supply from battery input;
  - outputs can drive resistive and inductive loads with 500 mA of current consumption and 1 A of current limitation;
  - current monitoring by analog current sense (accuracy: max 15% of read value in 50 mA 500 mA range);
  - all drivers are integrated in only one HSD with current sensing diagnostic, current limitation, overvoltage protection and thermal shutdown.

3A high side output

- 2 high side output 3 A (pin 19 and 26 of CN2):
  - On/off mode;
  - iLoad: 3 A;
  - Imax: 5.1 A;
  - minimum load resistance (Rload): min 3 Ω at 14 Vdc (5 Ω at 16 Vdc);
  - power supply from battery input;
  - outputs can drive resistive and inductive loads with 3 A of current consumption and 5.1 A of current limitation;
  - current monitoring by analog current sense (accuracy: max 10% of read value in 100 mA- 400 mA range and max 5% of read value in 0.5 A - 3 A range);
  - all drivers are integrated in only one HSD with current sensing diagnostic, current limitation, overvoltage protection and thermal shutdown.

LOAD DUMP PROTECTION

For the 12 Vdc system load dump not suppressed, internal resistance Ri = 2Ω generator.
Any other type of load dump protection, different from tests specified above, has to be evaluated by MTA.

TELLTALES

All 25 telltales are driven by the microcontroller.
L15 can also be driven directly by digital inputs pin 26 of CN1 with a high signal (5Vdc min.).
Each telltale is made by high efficiency monochrome (white) LED, PWM controlled to achieve light intensity compensation.

CONNECTORS

N. 1 Tyco 26 pin connector 3 keys (CN1), PN# 6437288-6.
N. 1 Tyco 26 pin connector 5 keys (CN2), PN# 6473418-2.



PINOUTS

CN1 (3 keys)

PIN	NAME	TYPE
1	IN_1	Digital input low side
2	IN_2	Digital input low side
3	IN_3	Digital input low side
4	IN_4	Digital input low side
5	IN_5	Digital input low side
6	IN_6	Digital input low side
7	IN_7	Digital input low side
8	GND (terminal 31)	Battery ground
9	IN_8	Digital input low side
10	IN_9	Digital input low side
11	IN_10	Digital input low side
12	IN_11	Digital input low side
13	IN_12	Digital input low side
14	VCTRL (terminal 15)	Ignition input
15	VB (terminal 30)	Battery power supply
16	IN_13	Digital input low side
17	IN_14	Digital input low side
18	IN_15	Digital input low side
19	IN_16	Digital input low side
20	D+ alternator	Alternator input
21	VB (terminal 30)	Battery power supply
22	IN_17	Digital input low side – wake-up
23	IN_18	Digital input low side
24	IN_19	Digital input low side
25	IN_20	Digital input low side
26	IN_21	Digital input high side – direct telltale

CN2 (5 keys)

PIN	NAME	TYPE
1	LIN	LIN
2	CAN1_H	CAN1_H (*)
3	CAN1_L	CAN1_L (*)
4	CAN1_T	CAN1_T
5	OUT_1	Digital output high side – 500mA
6	OUT_2	Digital output high side – 500mA
7	IN_22	Digital input high side
8	IN_23	Digital input high side – wake-up
9	IN_24	Digital input high side – wake-up
10	IN_25	Digital input high side
11	IN_26	Digital input high side
12	IN_27	Digital input high side
13	GND (terminal 31)	Battery ground
14	IN_35	Analog input
15	CAN2_H	CAN2_H
16	CAN2_L	CAN2_L
17	IN_	Digital input high side
18	IN_33	Analog input
19	OUT_3	Digital output high side – 3A
20	IN_34	Analog input
21	IN_32	Analog input
22	IN_28	Tristate input
23	IN_29	Tristate input
24	IN_30	Frequency input
25	IN_31	Frequency input
26	OUT_4	Digital output high side – 3A

(*) The software download must be done via CAN1 only.

LAYOUT

All the PN#s of Tellus dashboard share the same mechanical parts (the image on the TFT is only an example):



LCD

Features:

- TFT colors 3.5", 4:3 diagonal, QVGA;
- Normally black, transmissive, portrait;
- Active area: 53,28 mm (H) x 71,04 mm (V);
- 240 x RGB x 320;
- Operating temperature: -30 / +85 °C;
- Storage temperature: -40 / +90 °C;
- Luminance: 400 cd/m² Min, 600 cd/m² Typ.;
- Contrast ratio: 700 Min, 900 Typ.;
- Viewing angle: 80/80/80/80°;
- Antiglare treatment.

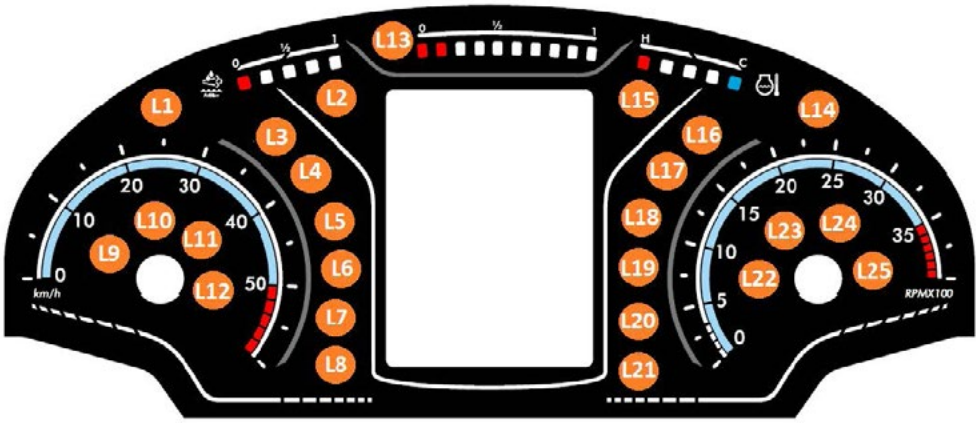
A software derating strategy has been implemented to preserve the display from overheating; contact MTA for details.

The image on the TFT is only an example.



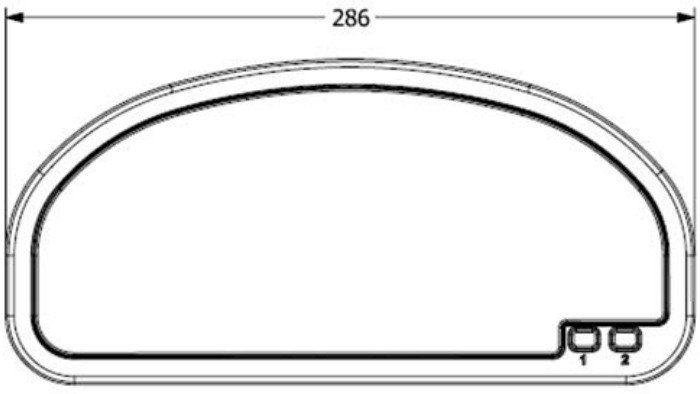
DIAL PLATE AND TELLTALES

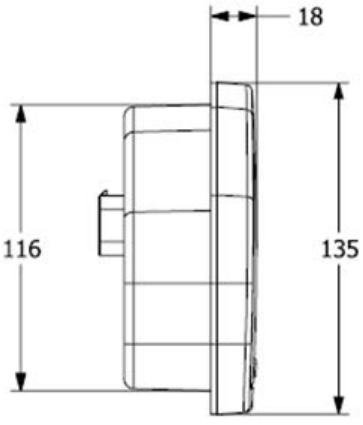
The dial plates here described are for reference purpose and available for the PN# of this document. Please refer to customer drawings for PN#s derived by this one (MTA off-the-shelf cluster), in which a custom dial plate could be included.

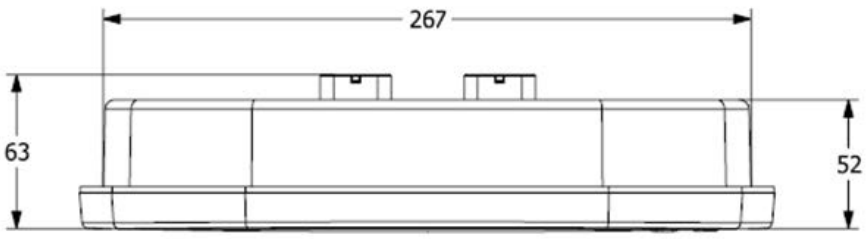


DIMENSIONS

The external dimensions are: 286 x 135 x 52 mm, they are common to all the product variants of the Tellus family.





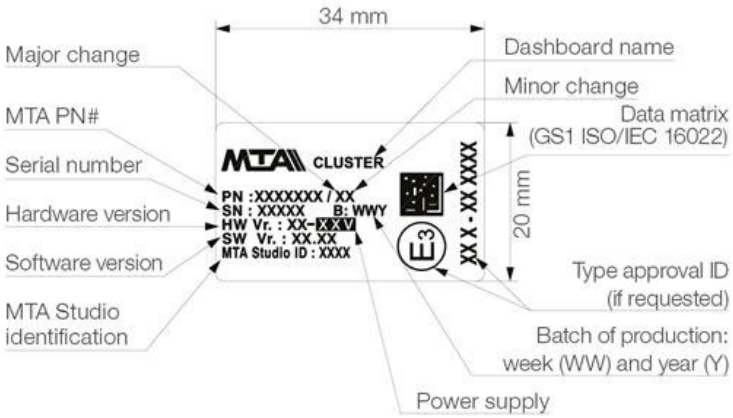


FIXING

Fixing is made by 2 spring clips.
Coupling with dashboard is done by rubber gasket.



LABEL DETAILS



PRODUCT QUALIFICATION

MTA reference test specification for Tellus dashboard is TS50_16 (rev. 6) which includes:

- environmental tests;
- mechanical tests;
- chemical tests;
- electrical tests;
- EMC tests;
- CE type approval.

Please contact MTA for any details.

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